MILITARY SPECIFICATION

CARTRIDGE, 30MM, TARGET PRACTICE, M788

This specification is approved for use by the U.S. Army Armament Research and Develpment Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and packaging of one type of electric primed cartridges for use in guns, 30MM, automatic.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	-	Fuze & Fuze Components, Environmental and Performance Tests for
MIL-STD-414	-	Samples procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-810	-	Environmental Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MIL-STD-1167 - Ammunition Data Card

MIL-STD-1168 - Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein:

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND ARRADCOM

PRODUCT AND PACKING DRAWINGS

11825901	Case Cartridge Assembly
11825947	Cartridge, 30MM, Target Practice, M788
11825892	Linked packing - Left Hand
11825891	Linked Packing - Right Hand
11825890	Bulk Packing
10542565	Shipping and Storage Container M592
11825879	Link
11825893	Marking, Shipping and Storage Container
11825969	Test Barrel, 30MM

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Code of Federal Regulations

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General.

3.1.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.1.2 Cartridge. The cartridge shall comply with all requirements specified on Drawing (Dwg. 11825947), all associated drawing, and with all requirements specified in applicable specifications.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the Contracting Officer and no deviation from that process shall be made without prior approval (ref 6.3).

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract-.

3.4 Projectile extraction. The projectile, when subjected to a bullet pull using a chamber simulating ring at a rate of 25mm per minute ±10 percent, shall have a retention force such that the mean value less-three times the standard deviation shall be greater than 8 kiloNewtons.

3.5 Electrical resistance, primed case. The electrical resistance of the primed case shall be as specified on Dwg. 11825901.

3.6 Propellant charge. The propellant charge used in each cartridge shall be maintained to meet the ballistic requirements-

3.7 Muzzle velocity. The projectile shall have a mean muzzle velocity of 795-815 meters per second (m/s) at 21 ± 5°c. The standard deviation shall not exceed 15 m/s at temperatures of -54°c, 21°c, and 71°c. following environmental exposure, the mean muzzle velocity shall not increase or decrease from the mean muzzle/velocity at 21 ±5°c by more than 30 m/s.

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3.8 Case mouth pressure. The mean peak pressure shall not exceed 270-310 MPa at 21°C ±5°C. The mean peak pressure shall not exceed 415 MPa at temperatures of -54°C and 71°C respectively. Following environmental exposure, the mean peak pressure shall not increase or decrease from the mean peak pressure at 21 ±5°C by more than 60 MPa

3.9 Environmental conditions. The cartridge shall not be damaged, deteriorated, or performance degraded beyond the limits of this specification after exposure to the following environmental conditions.

3.9.1 Temperature, humidity and altitude. Temperature/ humidity/altitude cycling conditions in accordance with MIL-STD-810, method 504, with the temperature ranging from -54°C to 65°C, relative humidity up to 95 percent, and pressure conditions equivalent to 15.2 kilometers altitude.

3.9.2 Salt fog. Exposureto salt c sea atmosphere specified in MIL-STD-331, Method 107.

3.9.3 Aircraft vibration. Vibration as specified in MIL-STD-81O, Method 5: 4 except as modified in Section 4.

3.9.4 Waterproofness. Immersion for one hour in water under a pressure of 0.1 MPa ± 10% 21°C.

3.1O Accuracy. The cartridge when fired in a single shot test barrel (Dwg 11825969} shall be such that the standard deviation of the impact points about the mean impact point shall not exceed 1.35 mil.

3.11 Action time. The mean action time plus 3 standard deviations shall not exceed 4.0 milliseconds (ms) at -54°C, 21°C and 71°C. The maximum individual action time shall not exceed 4.0 ms at -54°C 21°C and 71°C. The action time is defined as the time period between the application of electric current to the primer and the exiting of the projectile from the muzzle.

3.12 Function and casualty. The cartridge shall function as required and there shall be no separation or breakup of cartridge parts during the firing sequence. When fired in the automatic gun there shall be no gun stoppages caused by the ammunition. See 6.13.

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3.13 Workmanship l. The requirement for workmanship are as specified on the appl. 1 cable drawings f referenced specifications and the following:

3. 13.1 Process de fects. The cartridges shall be free of defects suck, as folds, wrinkles, buckles , splits , severe scratches , cuts, dents, pe~forat~ons and other forms of substandard workmanship.

3.13.2 Cleanliness. The car ridge and its components shall be free of all foreign matter (see 6. 14).

4. QUAITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First Article Inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I.

4.3.2.1 First article inspection. Prior to submission of the first article sample to the Government, the contractor shall inspect all components and assemblies for all contract drawings, and specification requirements.

4.3.3 Rejection. See MIL-A-48078 and the following:

4.3.3.1 First article sample failure. Failure of the sample to comply with the requirement of the drawings and specifications Shall. resuit in sample disapproval.

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TABLE I.First article tests.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Cartridge, 30MM, TP M788		SHEET 1 OF 1		DRAWING NUMBER: 11825947
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
See 1/ A	Projectile extraction	20		3.4	4.5.11
A	Propellant charge	20		3.6	4.5.12
A	Propellant contamination	20	Acc-0 Rej-1	3.13.2	
B	Velocity 2/	60		3.7	4.5.2
B	Chamber pressure 2/	60		3.8	4.5.3
B	Action time 2/	60		3.11	4.5.4
B	Accuracy 2/	60		3.10	4.5.13
B	Function & Casualty (MANN Barrel)	60	Acc-0 Rej-1	3.12	4.5.5
	Temperature/humidity/altitude	20	Acc-0 Rej-1	3.9.1	4.5.9.1
	Salt Spray	20	Acc-0 Rej-1	3.9.2	4.5.9.2
	Aircraft vibration	20	Acc-0 Rej-1	3.9.3	4.5.9.3
	Waterproofness	20	Acc-0 Rej-1	3.9.4	4.5.9.4
	Function & Casualty (Auto gun) 3/	200	Acc-0 Rej-1	3.12	4.5.5

NOTES: 1/ Tests identified by the same letter shall be conducted on the same test item. 2/ 40 cartridges at 21°C and 10 cartridges each at -54°C & 71°C. Data collected at -54°C and 71°C will be recorded for information only. 3/ Gun specified in the contract or purchase order.

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4.4 Quality conformance inspection.

4. 4. 1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Cartridge easer from. one interfix number from *one* manufacturer .

b. Projectile parts from one interfix lot number from one manufacturer

c. Primers from one inter fix lot number from one manufacturer.

d. Propellant from one lot number from one manufacturer .

e. Loaded cartridge case assemblies from one inter fix lot number from one manufacturer.

4 .4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MOD-C-63980(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, 30MM, TP M788 1/	1 OF 3	11825947
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major					
101	Missing case to projectile crimp	All	100%	3.1.2	Visual
102	Split or perforated case	All	100%	3.1.2	Visual
103	Exterior profile (max) 3/	All	100% 2/	3.1.2	Gage
104	Overall length (max)	All	100% 2/	3.1.2	Gage
105	Case crimp diameter (max) 4/	All	100% 2/	3.1.2	Gage
106	Propellant charge (min) 5/	All	100%	3.1.2	Gage
107	Max gap case-projectile interface	4.4.2a	.25%	3.1.2	Gage
108	Case projectile diameter between crimps 6/	4.4.2a	.25%	3.1.2	Gage, Visual
Minor					
201	Protective coating damaged or incomplete 7/	4.4.2a	.65%	3.1.2	Visual
202	Incorrect type cartridge	4.4.2a	.65%	3.1.2	Visual
203	Foreign material (except corrosion)	4.4.2a	.65%	3.1.2	Visual
204	Workmanship	4.4.2a	.65%	3.1.2	Visual

NOTES:

See Notes following 4.4.2.5.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, 30MM, TP M788	2 OF 3	11825947
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
205	Folds, wrinkles, buckles, deep scratches, cuts and dents	4.4.2a	0.65%	3.1.2	Visual
206	Corrosion	4.4.2a	0.65%	3.1.2	Visual
207	Marking incorrect, illegible or missing	4.4.2a	0.65%	3.1.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63980(AR)

PARAGRAPH 4.4.2.1	TITLE Cartridge, 30MM, TP M788		SHEET 3 OF 3		DRAWING NUMBER 11825947 / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TESTS				
Major:					
109	Projectile extraction *	Table II	Acc-0 Rej-1	3.4	4.5.11
110	Presence of waterproof seal (Projectile to case)				
111	Propellant contamination *		Acc-0 Rej-1		
112	Velocity **	Table III & IV		3.7	4.5.2
113	Pressure **			3.8	4.5.3
114	Action time **			3.11	4.5.4
115	Accuracy **			3.10	4.5.13
116	Function & Casualty ** (Mann Barrel)		Table VI	3.12	4.5.5 Table V

NOTES:
* Use same Test items for Majors 109, 110, 111.
** Use same Test items for Major 112, 113, 114, 115, & 116.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63980

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825903
4.4.2.2	Flashtube Assembly				NEXT HIGHER ASSEMBLY 11825901
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major:					
101	Ignitor material missing or incorrect (Broken pellets permissible)		.25	3.1.2	Visual
Minor:					
201	Evidence of poor workmanship		.65	3.1.2	Visual

NOTES:

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CLASSIFICATION Of DEFECT'S & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 11825901
4.4.2.3	Case Assembly		NEXT HIGHER ASSEMBLY 11825947

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major:					
101	Depth of primer seating	All	100% 2/	3.1.2	Gage
102	Primer/Case Resistance	All	100% 2/	3.1.2	Gage
103	Presence of Igniter Tape Seal	All	100% 2/	3.1.2	Gage
104	Dimpling of Cartridge Case	All	100%	3.1.2	Visual
105	Loose Primer	4.4.2a	25%	3.1.2	Manual
106	Corrosion	4.4.2a	25%	3.1.2	Visual
Minor:					
201	Primer Waterproofing Missing or Incomplete	4.4.2a	65%	3.1.2	Visual
202	Incomplete or Missing Primer Crimp	4.4.2a	65%	3.1.2	Visual
203	Evidence of Poor Workmanship	4.4.2a	65%	3.1.2	Visual

NOTES: See Notes following 4.4.2.5.

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QUALITY CONFORMANCE INSPECTION

DOD-C-63890 (AR)

CLASSIFICATION OF DEFECTS & TEST

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825890 or 11825891 or 11825892
4.4.2.4	Container, Shipping and Storage, 30MM Packed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Link improper (anyone of the four (4) tabs of link not in groove; link inverted; link damaged; link distorted		100%	3.1.1	Visual
MAJOR					
101.	Cartridge inverted		.40%	3.1.1	Visual
102.	Incorrect number of cartridges		.40$	3.1.1	Visual
103.	Cushions, fillers, liners and separators missing		.40%	3.1.1	Visual
104.	Tubes missing		.40%	3.1.1	Visual
105.	Link orientation (LH-RH)		.40%	3.1.1	Visual
106	Contents loose		.40%	3.1.1	Visual
MINOR					
201.	Mettalic seal missing		.65%	3.1.1	Visual

NOTES:

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AUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-C-63890(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Marking on Shipping and storage Container M592, 30MM Cartridge				11825893
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:	None defined				
MINOR:					
201.	Marking misleading or undenfifiable		0.65%	3.1.1	Visual

NOTES:

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NOTES FOR PARAGRAPH 4.4.2.1 & 4.4.2.3

1/ Visual standards/machine masters established and approved by the Government as defined in 6.4, shall be used in evaluating cartridge case and cartridge visual defects. Major components shall have previously been accepted in accordance with the acceptance criteria defined in toe specification listed in the applicable drawing.

2/ Inspection verification of prior 100% inspections shall be performed per (a) by utilization of Government approved machine masters or (b) MIL-STD-105, AQL = 0.25, to determine sample size minimum sample size = 315, zero defects allowed.

3/ If a chamber gage is used, the maximum force to insert the cartridge into the gage shall comply with either of the following criteria:

a. Apply a static force of not to exceed 270 Newtons.

b. Apply a dynamic force by hand (no mechanical aid) which does not damage the cartridge.

4/ The lot shall not be penalized during inspection verification for oversize conditions provided that:

a. The crimp is visually complete for 360° and,

b. A sample quantity of the discrepant cartridges in accordance with Table II meet the projectile extraction requirement of 3.4

5/ A cartridge weighing less than the value determined by the formula specified in 6.8 shall be rejected. The 100% inspection method may be by verification of weight of each cartridge or by probe for a fixed minimum propellant height in a charged cartridge case. If the probe method is used, the followinq method shall be used to establish the minimum height:

a. Establish the mean charge weight in accordance with 4.5.12.

b. Charge a minimum of 25 cases to the specified weight, measure each case and calculate the mean propellant height.

c. Repeat step b with a propellant charge weight 1 gram less less than the established mean charge weight.

d. Adjust the probe height such that any cartridge in which the propellant charge is 1 gram below the mean weight will be rejected.

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6/ A gap of 0.25mm or less at any point in the circumference is acceptable.

7/ The lot shall not be penalized for anodize crazing in the cartridge case crimp grooves (see 6.9).

TABLE II. Sample size for projectile extraction test.

Lot Size	Sample Size
0 - 800	10
801 - 1, 300	15
1,301 - 3,200	20
3,201 - 8,000	25
8,001 - 22,000	30
22,001 and over	35

Reference: MIL-STD-414, Level 2, Table A-2
Except for lots of less than 800 units.

TABLE III. Sample size for 21°C ballistic tests.

Lot Size	First Sample Size	Second Sample Size
0 - 800	30	35
801 - 1,300	35	40
1,301 - 3,200	40	50
3,201 - 8,000	50	75
8,001 - 22,000	75	100
22,001 - 110,000	100	140
110,000 - 550,000	150	190
Greater than 550,000	200	255

Reference MIL-STD-414, Level 4, Table A-2
Except for lots of less than 800 units.

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TABLE IV. Sample size for 71°C and -54°C ballistic tests.

Delivered Lot Size	Sample Size for Each Temperature: First Sample Size	Second Sample Size
0 - 800	10	10
801 - 1,300	10	10
1,301 - 3,200	10	15
3,201 - 8,000	15	20
8,001 - 22,000	20	20
Greater than 22,000	25	25

Reference: MIL-STD-414, Level 1, Table A-2
Except for lots of less than 3200 units.

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TABLE V. Firing defects 1/

Item	Defect	AQL
Cartridge	Misfire due to cartrige 2/	0.065
	Failure to than chamber due to cartridge 3/	0.065
	Failure to extract due to cartridge 4/	0.065
	Projectile remaining in bore	0.065
Primer	Blown primer 5/	0.25
	Loose primer 6/	0.65
	Primer leak 7/	0.65
Cartridge Case	Split or rupture in the cartridge case 8/	0.40
	Detached metal from the cartridge case 9/	0.25
	Erosion, internal or external 10/	0.40
	Burn-through, base 11/	0.065
	Burn-through, major 11/	0.40
	Burn-through, minor 11/	0.65
Projectile	Metal parts separation 12/	0.065

1/ These defects apply only to cartridges fired in the approved test weapon.

2/ Misfire. If a misfire occurs, a second firing voltage application is not allowed unless a non-ammunition cause for the misfire is confirmed and corrected.

3/ Failure to chamber. The cartridge cannot be inserted into chamber sufficient distance to allow the breach to be engaged or cartridge cannot be bottomed by the breach within the limits of an axial load of 270 Newtons.

4/ Failure to extract. After firing, the cartridge case cannot be removed without failure of the extractor rinq. Foreign debris at cartridge/chamber interface causing cartridge case removal difficulty does not constitute a valid failure.

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5/ Blown primer. The cartridge case shall be visually examined. A blown primer is a prime which when the cartridge is fired, is separated completely from the head of the cartridge case, and both the head of the case and the primer pocket are enlarge, and deformed.

6/ Loose primer. The cartridge case shall be visually examined. A loose primer is defined as a looseness which permits the fired primer to fall from the primer pocket after the cartridge has been fired, without distortion of the primer pocket.

7/ Primer leaks. Primer leads shall be defined as follows:

a. Suspect. A suspect primer leak is defined as evidence of smudging or discoloration around any part of the periphery of the primer cup, button, insulator, or primer pocket. The fired primer must be removed from the case and disassembled to determine whether actual gas leakage has occurred.

b. Actual. A primer leak suspect, which upon disassembly, shown erosion of the primer insulator, or which shows a definite leakage path such as a crack or split.

c. A dislodged primer button is not cause for rejection.

8/ Split or ruptured cartridge case. The carrtridge case shall. be visually examined. A split or rupture is acceptable providing proper obturation is obtained.

9/ Detached metal. The cartridge case shall be visually examined. Metal sheared or missing from the exterior profile of a fired cartridge case, such as a rim or neck shear or ignition system component, shall be classed as detached metal. Shavings or metal from the interior wall of the neck in the crimp area shall not be considered as detached metal.

10/ Mouth erosion. Cartridge case mouth erosion is defined as erosion of an area qreater than 160 square millimeters to a depth such that the remaining thickness of the cartridge case mouth is reduced to less than 0.25mm.

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11/ Burn - through.

Major. Major burn-through is defined as a hole through the case due to melting which exceeds 2.5mm in any dimension.

Minor. Minor burn-through is defined as a hole through the case due to melting which is 2.5mm or less in any dimension.

Base. Base burn-through is defined as a hole in the base area not to include primer leaks. The base area is defined as the area of the case not supported by the chamber.

12/ Loss of projectile structual integrity (metal parts only). See 4.5.8.

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TABLE VI. Function and casualty firing defect accept\ reject criteria

Lot Size	AQL			
	0.065	0.25	0.40	0.65
501 - 800	0/1*	0/1*	0/1*	0/1*
801 - 1,300	0/1*	0/1*	0/1*	0/2 1/2
1,301 - 3,200	0/1*	0/1*	0/2 1/2	0/2 2/3
3,201 - 8,000	0/1*	0/1*	0/2 2/3	0/3 4/5
8,001 - 22,000	0/1*	0/2 2/3	0/3 3/4	1/3 4/5
22,001 - 1100,000	0/1*	0/2 2/3	0/3 4/5	1/4 6/7
110,001 - 550,000	0/1*	0/3 3/4	1/4 4/5	2/5 6/7
Over 550,000	0/1*	1/3 3/4	2/4 6/7	3/6 8/9

* Indicates single sampling plan, i.e., no retest

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4.5 Test methods and procedures.

4.5.1 Unlisted firing defects. The lot should be suspended and referred to the contracting officer for disposition if any malfunction or casualty not covered by this specification occurring in any firing test.

4.5.2 Muzzle velocity. Velocity shall be measured by the use of velocity coils or light screens. The test cartridges shall be conditioned at -54°C, 21°C and 71°C respectively before firing. The mean velocity and standard deviations from the mean of the sample cartridges in each group, shall conform to 3.7. The velocity shall be corrected to the muzzle velocity as follows:

$$V_m = \frac{d}{t} 1 + c \quad d_2 + 0.5d_1$$

where:

t = time between screens, seconds

d_1 = distance between screens, meters

d_2 = distance from muzzle to first screen, meters

c = constant of proportionality,0.75 meters/second/meter

4.5.3 Case mouth pressure. The test rounds shall be conditioned at -54°C, 21°C and 71°C temperatures respectively and fired from a test fixture barrel. Pressure measurements shall be made at the case mouth with the piezo-electric transducer. A permanent record shall be made of the resulting pressure. Data at -54°C and 71°C will be recorded for information only. This record may be a pressure trace, paper print out of the data or other similar instrument reading. The pressure shall conform to the requirements of 3.8.

4.5.4 Action time. The test rounds shall be conditioned at -54°C,21°C and 71°C and fired from a test fixture barrel. The time between application of voltage to primer and exit of the projectile from the muzzle shall be measured. The time shall comply with the requirement of 3.11. Any corrected action time of 2.40 milliseconds or less shall be disregarded as an instrumentation error.

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4.5.5 Function and casualty. These tests shall be conducted, using the test methods prescribed in AMCR-715-505, Volume 8, Chapter 1, Section 3 as a guide. Witness panels for structural integrity shall be placed approximately 5 meters from the muzzle. A failure is characterized by observance of any of the firing defects listed in Table V.

4.5.6 Test barrel. The test samples shall be fired from a test barrel conforming to Drawing 11825969. See 6.6 for definition of acceptable test fixture barrel for l test purposes.

4.5.7 Automatic gun. Test rounds, when required, shall be fired in the M230 Gun. Test results shall conform to the requirements of 3.12.

4.5.8 Projectile structural integrity. Projectile structural integrity shall be verifled in conjunction with the ballistic tests of 4.5.2, 4.5.3 and 4.5.4. A witness board/card shall be located approximately 5 meters from the gun muzzle. Loss of projectile structural integrity is defined as projectile fracture, break-up, rotating band failure as defined in 6.1O, or loss of parts as evidenced by microflash photography and/or hole(s) in the witness board or recovery of fragments.

4.5.9 Environmental tests. Unless otherwise specified, performance after environmental exposure, as specified in the applicable test procedure, shall consist of conducting the tests of 4.5.2, 4.5.3 and 4.5.4, except all test rounds shall be conditioned at 21°C prior to firing.

4.5.9.1 Temperature, humidity and altitude. The cartridge shall be subjected to the temperature/humidity/altitude test of MIL-STD-81O, Method 504. After exposure, the test item shall be examined for deterioration and subjected to the tests referenced in 4.5.9.

4.5.9.2 Salt fog. The cartridge shall be subjected to the salt fog test of MIL-STD-331, Method 107 for a period of 48 hours . After exposure, the test item shall be subjected to the tests referenced in 4.5.9.

4.5.9.3 Aircraft vibration. The ammunition shall be subjected to a broad band random vibration profile in accordance with the requirements of MIL-STD-810, Method 514.2, procedure I, Category C. The test will be conducted with an AAH magazine. The vibration spectrum shall be as shown in Figure 2. The test item shall be mounted to a test fixture sirnd.sting a feed system in a normal airplane mounting. The test duration shall be 45 minutes in each of three mutually perpendicular axis at each of two temperature conditions, -40° ±2.8°C and 65.5 ±2.8°C. Following exposure, the test items shall be subjected to the tests referenced in 4.5.9.

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4.5.9.4 Waterproofness. Sample cartridges shall be immersed for one hour in water under a pressure of 0.1 MPa ±10% gauge at 21°C. The cartridges shall be taken out and wiped dry. The test quantity shall then be tested in accordance with 4.5.9 to verify conformance with the requirements of 3.9.

4.5.10 Primed cartridge cases. The primed cartridge case resistance shall meetthe requirements of drawing 11825901.

4.5.11 Projectile extraction. The base of the cartridge case shall be secured in a holding fixture and a ring simulating the gun chamber shall be installed in the case neck area. A clamp shall be applied to the projectile so that a tensile load may be applied along the longitudinal axis of the round. An axial load shall be applied at a constant strain rate of 25MM per minute ±10% until the projectile is withdrawn from the cartridge case. The peak axial load shall be recorded and compared to the requirements of 3.4. A case neck separation during this test shall be cause for rejection. A split of greater than 5MM observed during projectile extraction shall be cuase for rejection.

4.5.12 Propellant charge The charge of propellant from a given lot, which is required to meet the performance requirements of 3.7, 3.8 and 3.11 shall be determined by ballistic tests. The minimum charge weight assigned shall be used in determining the minimum cartridge weight in accordance with 6.8.

4.5.13 Accuracy. The test shall be conducted using a single shot test barrel with cartridge conditioned to 21°C. The target shall be positioned at approximately 25 meters from the muzzle. After three barrel fouling shots have been fired the test snots shall be fired without moving the gun or target. The target impacts shall be measured for dispersion from the calculated center of impact as described in AMCR 715-505, Vol 8, Chapter I, Section 4.4. The impact dispersion of these shots wll comply with the requirements of 3.10. Testing shall not be conducted if wind velocity including gusts, is 10 knots or more.

4.5.14 Inspection Equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the “Paragraph Referernce/Inspecion Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract See Section 6 of MIL-A-48078 and 6.15 herein

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4.5.15 Reference rounds. Reference rounds shall be fired in conjunction with ballistic and accuracy tests. Unless otherwise specified, the reference rounds shall be assembled in accordance with the requirements of Drawing 11825947 and inspected in accordance with this specification with the charge weight controlled to ± 0.2 grams.

4.5.16 Ballistic reference rounds. A minimum of 10 reference rounds shall be fired during each ballistic firing test and the mean muzzle velocity, case mouth pressure and action time recorded. Any individual round readings which vary from the mean by greater than 12 meters per second, 20 MPa or 0.3 milliseconds respectively, shall be disregarded and a new mean velue shall be computed. If more than three rounds are disregarded, additional reference rounds shall be fired, provided they are fired during the same day as the applicable tests are conducted. The differences between the mean reference round test values and the reference acceptance baseline values shall not exceed 10 meters per second, 23 MPa, or 0.35 millisecond. Reference round data shall be utilized as defined in 6.20.

4.5.17 Accuracy reference rounds. The accuracy reference baseline shall be established during lot acceptance testing of the reference lots in accordance with the tests of Table 11, except a minimum of 60 rounds, divided into 20 round groups,shall be used to establish the baseline. The accuracy of each 20 round group shall be determined and an average calculated for the baseline. Reference rounds shall be utilized as defined in 6.20.

5. PACKAGING

5.1 Packaging and packing. Cartridge shall be preserved, packaged and packed in acordance with the applicable requirements for the shipment of hazardous items as defined in Title 49, Code of Federal Regulations, Parts 100 through 199.

5.1.1 Packing.

5.1.1.1 Linked packing. The linked pack shall consist of linked cartridges packed in the M592 shipping and storage container, 10542565 and may include provisions for interconnecting ammunition belts. The links shall conform to Drawing 11825879. The pack shall also include provisions for a right and left hand pack. The packing ratio of left to right belts shall be specified in the contract or purchase order.

5.1.1.2 Bulk packing. The bulk pack shall consist of cartridges packed in he M592 shipping and storage container .

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5.1.2 Ammunition contianer interior packing.

5.1.2.1 Link packing. The interiopr packing shall be in accordance with Drawing 11825891 or 11825892 as applicable.

5.1.2.2 Bulk packing. The interior packing shall be in accordance with Drawing 11825890.

5.2 Markings Marking for shipment shall be in accordance with the instructions of the contract and shall comply with Dwq No. 11825893. In addition, exterior containers shall show the caution markings required by Title 49, Code of Federal Regulations, indicating the appropriate classification of ammunition.

5.3 Documentation with shipment. Each shipment of ammunition shall contain data cards in accordance with MIL-STD-1167, lot number assignment in accordance with MIL-STD-1168, date of manufacture and complete definition of the ammunition including such items as case, primer, flashtube assembly, propellant, charge weights, projectile and lot identification.

5.4 Shipping. When shipments of more than one lot are made at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent. mixing of the lots in transit.

6. NOTES

6.1 Intended use. Thi s specification covers assembling and packing for Cartridge, 30MM, Target Practice, M788 for firing in the M230 Gun, the Aden MX IV and DEFA series of guns.

6.2 Ordering data. See MIL-A-48078.

6.3 Process deviation. A process deviation defined as a change in the approved basic method of manufacture or an operational change which may alter the metallurgical or physical properties of the item.

6.4 Visual defects standards. Components and assemblies which have been rejected during initial production shall be segregated and visual standards established jointly by representatives of the Government and the contractor. Subsequent application of these standard for acceptance inspection shall be specified in the co-tract or purchase order.

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6.5 Accuracy measurement procedure.

a. Select a reference point on the target and establish X and Y Axis through this point. Measure X and Y distance to each impact point.

b. Calculate mean impact point $(\bar{X}, \bar{Y})$, (i.e., $\bar{X} = \Sigma X_i/N, Y = \Sigma Y_i/N$).

c. Calulate standard deviations for X and Y.

$$\sigma_x(m) = \sqrt{\frac{\Sigma (X_i - \bar{X})^2}{N-1}} \quad , \sigma_y(m) = \sqrt{\frac{\Sigma (Y_i - \bar{Y})^2}{N-1}}$$

d. Calculate radial standard deviation (σ_R) base on σ_x and σ_y

$$\sigma_R(m) = \frac{\sigma_x + \sigma_y}{2}$$

An alternate method of measuring accuracy is as follows:

$$RSD = \sqrt{(\sigma x)^2 + (\sigma y)^2}$$

NOTE : If this alternate method is used, all reference values for accuracy in this specification must be multiplied by 1.5.

e. Adjust calculation for 1000 m target.

$$\sigma_R(A\ 1000\ m) = \frac{1000}{(\text{range to target})}\ \sigma_R(m)$$

f. convert σ (at 1000 m) to roils dispersion)

1 mil = 1 m at 1000 m

Data to be provided shall be individual shot x and y impact location data and calculated dispersion standard deviations σx, σy, and σr.

6.6 Mann barrel. The barrel used for preproduction and lot acceptance testing shall not be used for testing if any of the following conditions exist.

a. The barrel does not meet drawing requirements.

b. Degradation of reference lot muzzle velocity by more than 15 meters per second.

If accuracy degradation in excess of 0.4 mil still exists, the barrel shall be replaced.

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6.7 Standard deviation. The standard deviation for use in testing for projectile extraction, muzzle velocity, chamber pressure and action time shall be calculated in accordance with the following formula:

$$\sigma = \sqrt{\frac{\sum (x_i - x)^2}{n-1}}$$

6.8 Minimum cartridge weight. The minimum cartridge weight shall be determined for each lot based on the minimum propellant charge weight selected per 4.5.12. The formula for minimum carttridge weight, when assembled in accordance with Drawing 11825947 is defined below:

Minimum weight (grams) = 292 grams + charge weight (grams)

6.9 Projectile coating defect criteria. Projectiles indicating exposure of bare metal or underlying phosphate or organic pre-treatment shall be considered unsatisfactory if any of the following conditions exist.

a. Any area exceeding 60 square millimeters.

b. More than one area exceeding 30 square millimeters.

c. More than five areas of 15 square millimeters or less.

6.10 Rotating band failure criteria. Rotating band failure shall be defined as loss of portion of the band which exceeds a 13MM length and has evidence of band seat serration imprint, or the wiping down to the bourellet diameter (no engraving in evidence) over the total length of the rotating band. For classification as a defect, there must be evidence, either by microflash photographs, recovery of the band or portions thereof, or by hole(s) in the witness board/screen. Normal band fringing is acceptable.

6.11 Muzzle velocity correction example. A velocity of 800.2 meters per second is recorded in a test Instrumentation configuration with the mid-point between the velocity coils located 30 meters from the muzzle. The muzzle velocity is calculated as follows:

Muzzle Velocity = 800.2 + (0.7S 30) = 823 meters per second

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6.12 Data reporting. Final data calculations shall be rounded off as follows:

Velocity - Nearest meter per second

Pressure - Nearest 0.1 MPa

Action time - Nearest 0.01 ms

Accuracy - Nearest 0.01 milliradian

6.13 Gun stoppage A gun stoppage is defined as failure to complete a singe round cycle.

6.14 Contamination of explosive components. Particular cafe should be exercised to aviod contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfire(s) or increase action time to make the ammunition unsuitable for use.

6.15 Submission of designs for approval See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAF-I, Dover, NJ 07801. The letter off submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.16 Retest procedures. Failure of the materiel sample to comply with the velocity pressure and action time requirements shall result in a retest for the failed requirements. Sample size shall be as indicated in the applicable table.

6.17 Accuracy retest criteria. If the Mann barrel dispersion standard devlation σ (test) exceeds the requirements specified in 3.2.1.5 the following procedure shall be followed:

a. Fire a 20 round accuracy test utilizing reference TP ammunition for which the Mann barrel dispersion baseline is known. Calculate the radial. standard deviation of the TP rounds ($\sigma_{TP_{ref}}$).

b. Calculate the difference between the TP reference baseline and a 20 round test group.

$$\sigma_{TP_{ref}} - \sigma_{TP_{baseline}} = \Delta\sigma_{TP}$$

c. Correct to standard deviation of the test ammunition as follows:

NOTE : $\sigma_{corrected} = \sigma_{test} - \Delta\sigma_{TP}$

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6.18 Abnormal test results (velocity pressure and action time). In the event of an individual abnormal test result, where one of the three critical parameters of muzzle velocity, chamber pressure, or action time was not obtained or is inconsistent with the other two and falls on or outside of the mean plus or minus four sigma value, (calculated without the individual abnormal test result) , the abnormal test result may be eliminated from the lot acceptance calculation for the tests of Table II. Calculation of the mean and standard deviation values shall be made both with and without the abnormal data and the specific abnormal data shall be noted . The criteria for acceptance of the lot shall be based on the data base without the abnormal data. If more than three rounds are disregarded additional rounds shall be fired, provided they are fired during the same day and under the same environment as the abnormal test result rounds.

6.19 Temperature tolerances. Unless otherwise specified all temperatures are tolerenced ± 5°C.

6.20 Calibration of instrumentation. A reference lot of 30MM TP ammunition will be manufactured. It will be tested and established as a control baseline. The reference lot accuracy will be established by averaging the first three-20 round accuracy tests. Prior to acceptance testing of each lot, a minimu of ten rounds per day will be fired to establish pressure, velocity and action time. Ballistic variation between reference and acceptance tests lots will be adjusted as follows:

$$X_a + (X_{ra} - X_r) = \text{corrected } X_a$$

Where X_a is mean of acceptance lot

X_r is mean of reference lot.

X_{ra} is mean of acceptance baseline.

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